Heartologies Design Studio



SHORE

MAGAZINE

DIVE IN

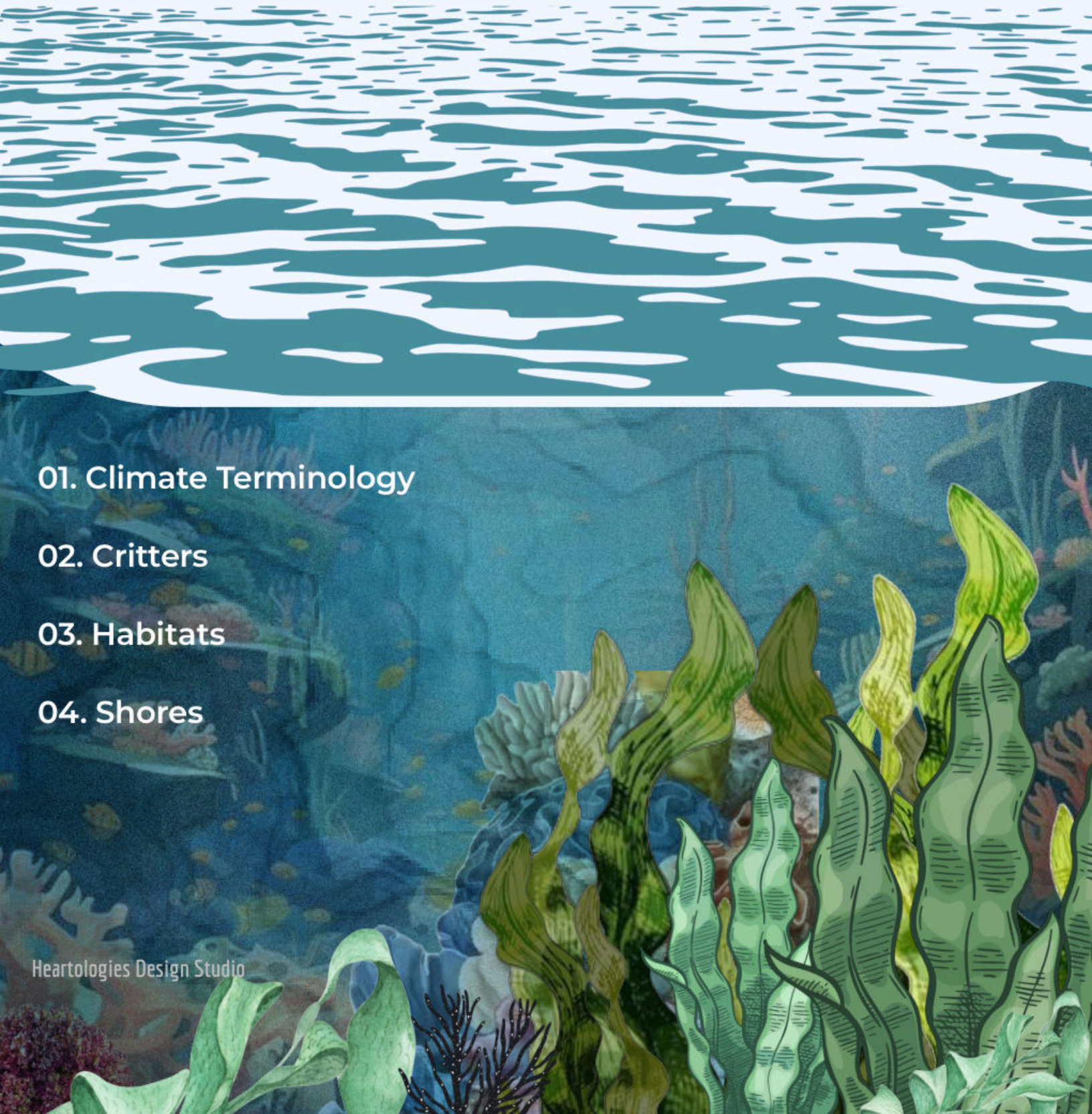
Learn more
Game Terminology

Let's take a closer look at the fascinating world of Shore Thing, where the shifting sands and evolving shorelines of barrier beaches come to life.

Heartologies Design Studio



DEEPER DIVE



01. Climate Terminology

02. Critters

03. Habitats

04. Shores

Heartologies Design Studio

MAGAZINE



Climate Terminology

Let's take a closer look at the fascinating world of Shore Thing, where the shifting sands and evolving shorelines of barrier beaches come to life.





An ice sheet or continental glacier is a large body of ice that covers at least 50,000 square kilometers of land all year round. Right now, there are two ice sheets on Earth: one covering Antarctica and one covering most of Greenland. If both ice sheets were to completely melt, global sea levels would rise a whopping 67 meters. At present, however, ice sheet melting contributes about 1 total millimeter of sea level rise per year, with 0.7 mm per year coming from Greenland and 0.3 mm per year coming from Antarctica.



ICE Sheets

Heartologies Design Studio

The melting of Greenland's ice sheet alone currently contributes 0.7 millimeters of sea level rise per year, impacting coastal areas along the Atlantic, including Rhode Island and Georgia. As sea levels rise, these regions face heightened risks of erosion, flooding, and habitat loss, intensifying the challenges for barrier beaches and coastal ecosystems in these areas.



MAGAZINE

Critters

Dive in and get to know the amazing coastal critters that call our shores home—each one plays a vital role in the beach ecosystem and needs our help to thrive!"



Heartologies Design Studio

The southern king crab, also known as the Patagonian king crab, thrives along the cold waters of southern Argentina, especially near Tierra del Fuego. These crabs are essential to the marine ecosystem, feeding on sea urchins, snails, and small fish, which helps balance local species populations. Known for their impressive size, with some reaching over 20 pounds, southern king crabs are also a valuable resource for fisheries in Argentina. However, climate change and overfishing have raised concerns about their habitat and long-term sustainability, making conservation efforts increasingly important for this remarkable species.

Southern King Crab



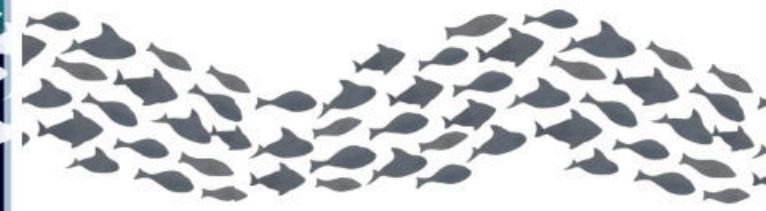
70% *of their diet made up of sea urchins, snails, and fish!*

Crabs are crustaceans that inhabit all of the world's oceans. They experience the effects of climate change in a variety of ways. Some crabs can live in warming waters but experience physical changes that require them to eat more in order to survive. Others are forced out of their homes by warming temperatures or the destruction of shoreline ecosystems.

MAGAZINE

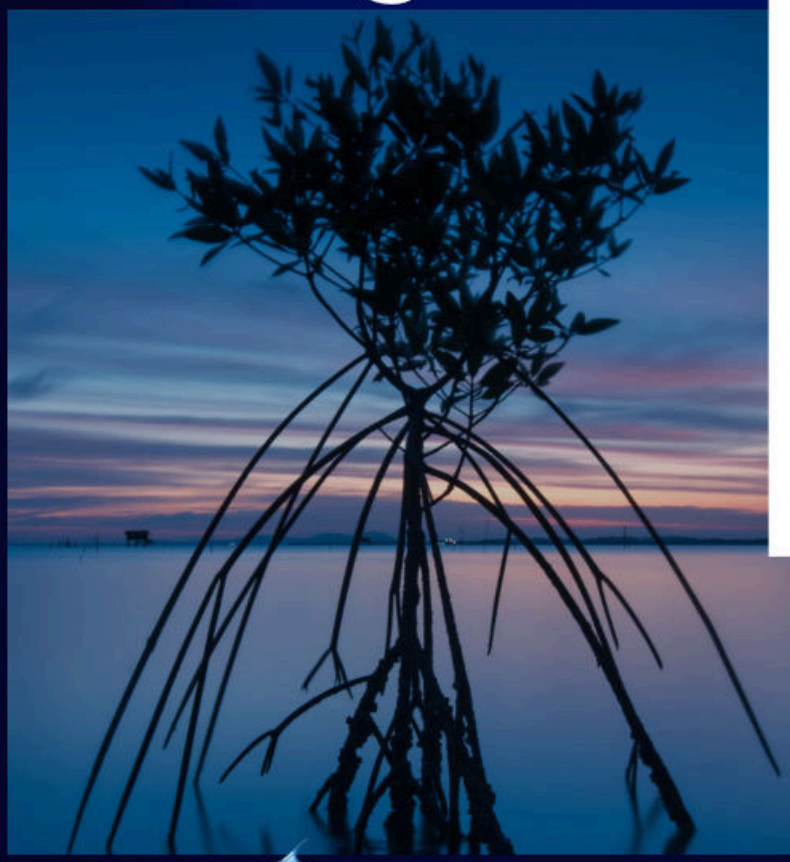
Habitats

Heartologies Design Studio



"Explore the vibrant natural habitats along our shores—each one teems with life and relies on our care to thrive!"

Mangroves

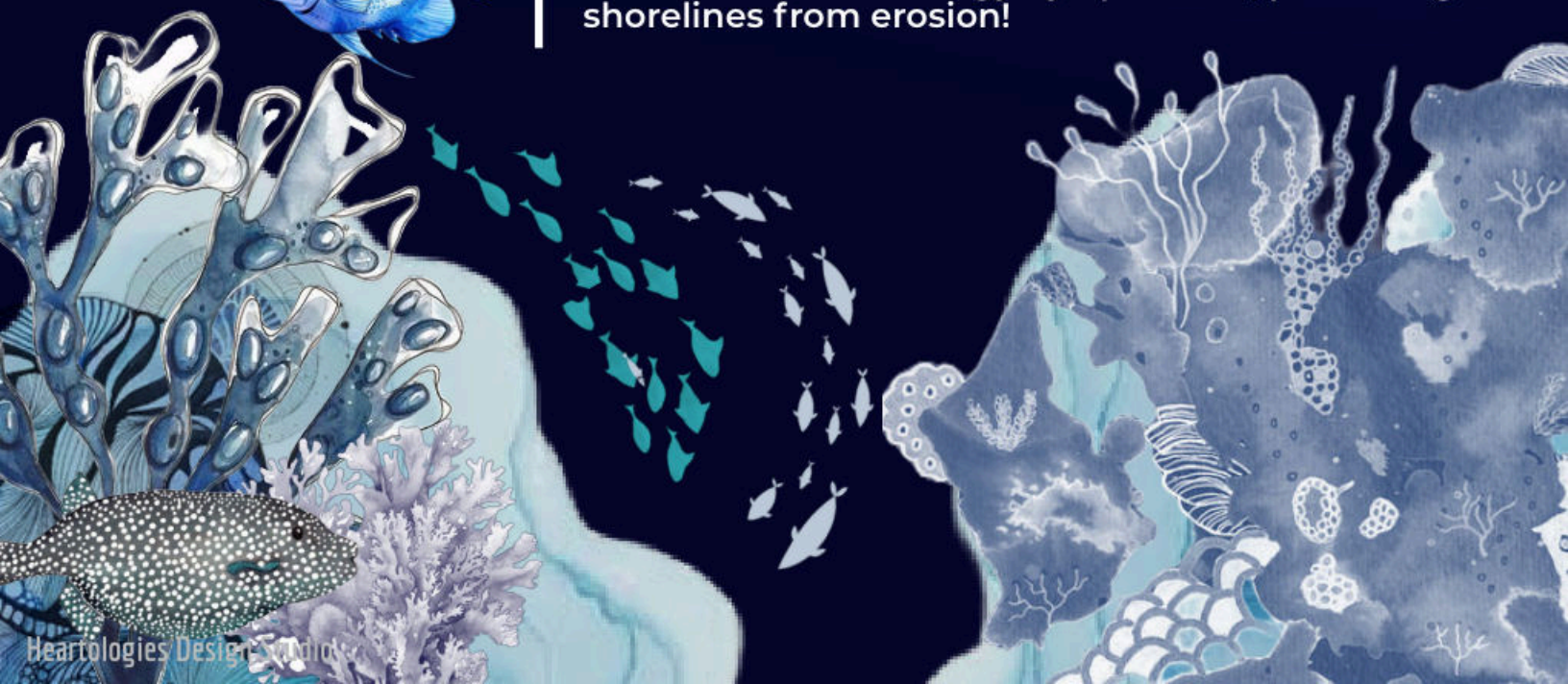


Mangroves are coastal forests made up of salt-tolerant trees and shrubs that thrive along shorelines in tropical and subtropical regions. These unique ecosystems are crucial habitats, providing shelter and breeding grounds for a variety of species, including fish, crabs, and birds. Mangroves' dense roots stabilize shorelines, reducing erosion by trapping sediments and buffering coastlines against powerful waves, especially during storms and hurricanes.

Mangroves can store up to five times more carbon per acre than tropical rainforests, helping to fight climate change!



Mangroves support over 300 species of fish and can reduce wave energy by up to 66%, protecting shorelines from erosion!



Heartologies Design Studio

MAGAZINE

Shores

"Step into the vibrant world of our shores—each stretch of sand and surf is alive with creatures and habitats that depend on our care to thrive."

Heartologies Design Studio

Heartologies Design Studio

Rhode Island Shores

known as the “Ocean State,” boasts an impressive 400 miles (640 km) of coastline along Narragansett Bay and Rhode Island coastal waters

The state’s beaches, like those in Misquamicut and Matunuck, serve as important buffers against storm surge and flooding, but they are increasingly vulnerable to erosion and loss. Coastal management practices in Rhode Island, such as dune restoration and artificial beach replenishment, are ongoing efforts to preserve these vital shorelines.

Misquamicut Beach: 41.3198° N, 71.8257° W
Matunuck Beach: 41.3734° N, 71.5437° W



Rhode Island’s coastline, particularly its barrier beaches along the southern shore, faces significant risks from rising sea levels and increasingly severe storms. These forces gradually push sand landward, reshaping beaches and threatening coastal communities and infrastructure





"The ocean stirs the heart, inspires the imagination, and brings eternal joy to the soul." **— Wyland, marine life artist and conservationist**

Illustrated By: Zoii Henry
Heartologies Design Studio